

Lab1-Posets_Intro

May 14, 2026

1 Posets in Sage - An Introduction

1.1 Math 737 - Lab 1

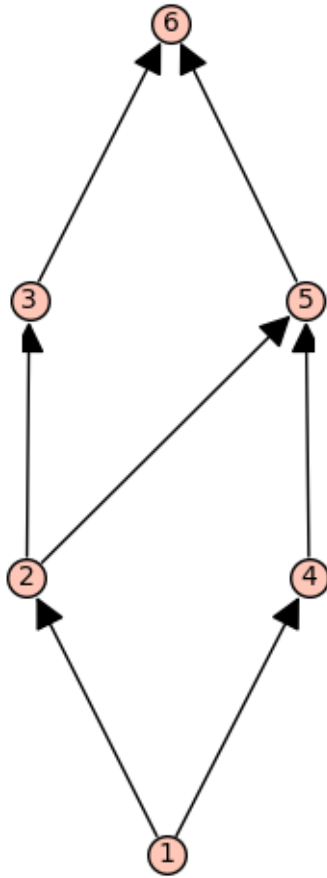
1.2 First define a poset.

1.2.1 Make a list of the elements of your poset, then a list of the covering relations between poset element.

1.2.2 Evaluate the cell below by putting your cursor in the block and then pressing shift + enter

```
[3]: elem = [1,2,3,4,5,6]
      rels = [[1,2],[2,3],[4,5],[5,6],[1,4],[2,5],[3,6]]
      Q = Poset((elem,rels))
      Q.plot()
```

[3]:

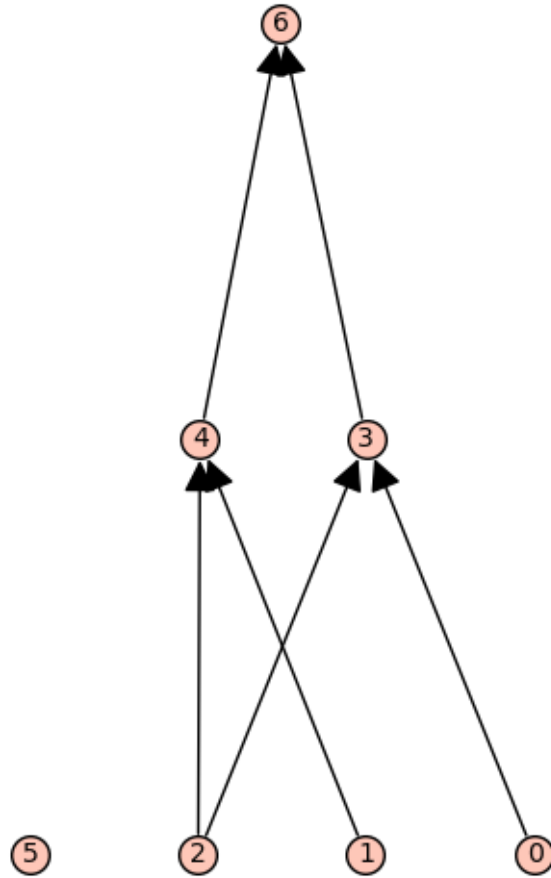


1.3 You can also make a random poset.

1.3.1 Evaluate the cell below multiple times to see multiple random posets of 7 elements.

```
[4]: myposet = Posets(7).random_element()  
plot(myposet)
```

[4]:



1.3.2 Exercise 1: Make some of the example posets P, Q, R, S, T, N, from our last class. We already made Q above.

```

[5]: # P
elem = ['a','b','c','d','e']
rels = [['a','b'],['b','d'],['d','e'],['a','c'],['c','e']]
P = Poset((elem,rels))
P.plot()

# Q
elem = ['a','b','c','d','e']
rels = [['a','b'],['a','c'],['b','d'],['c','d'],['c','e'],['d','f'],['e','f']]
Q = Poset((elem,rels))
Q.plot()

# R
elem = ['a','b','c']
rels = [['a','c'],['b','c']]
R = Poset((elem,rels))

```

```

R.plot()

# S
elem = ['a','b','c','d']
rels = [['a','c'], ['a','d'], ['b','c'], ['b','d']]
S = Poset((elem,rels))
S.plot()

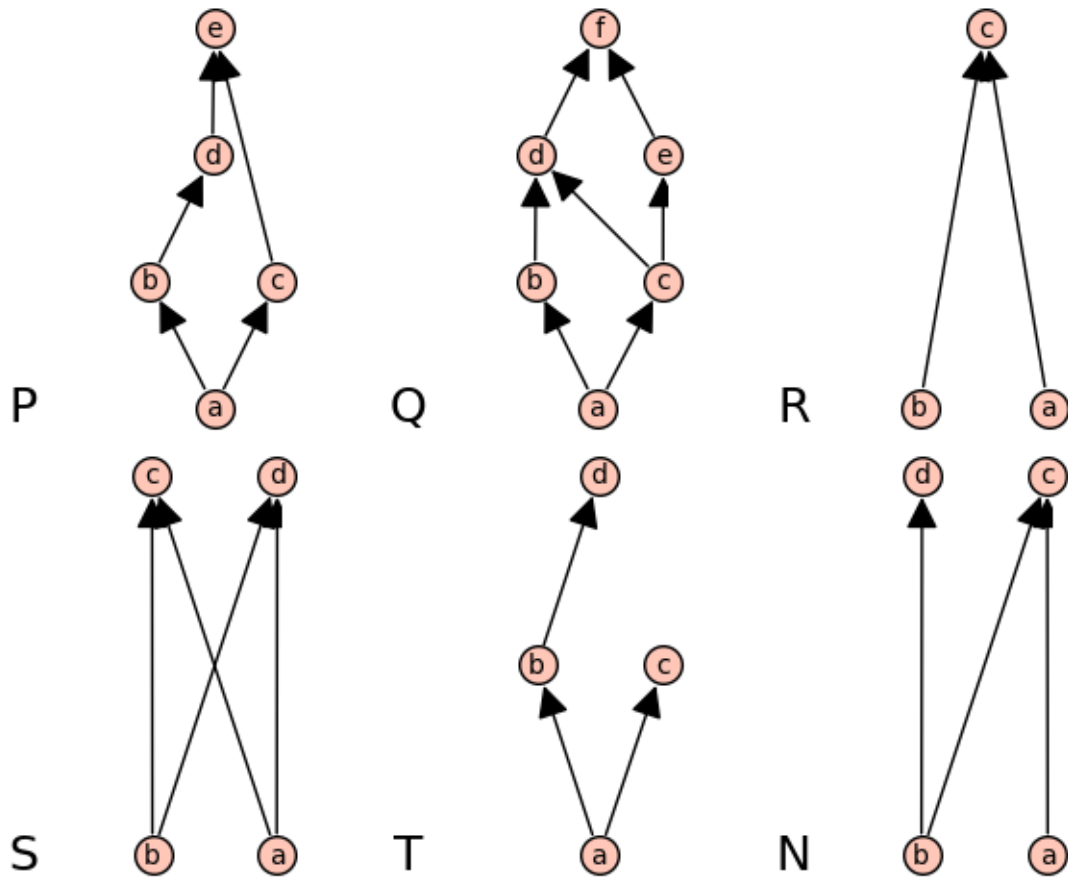
# T
elem = ['a','b','c','d']
rels = [['a','b'], ['b','d'], ['a','c']]
T = Poset((elem,rels))
T.plot()

# N
elem = ['a','b','c','d']
rels = [['a','c'], ['b','c'], ['b','d']]
N = Poset((elem,rels))
N.plot()

def L(X, name):
    g = X.plot(axes=False, figsize=4)
    return g + text(name, (0,0), fontsize=18, color='black')

graphics_array([
    [L(P, "P"), L(Q, "Q"), L(R, "R")],
    [L(S, "S"), L(T, "T"), L(N, "N")]
]).show()

```



1.3.3 Exercise 2: Use Sage to verify your answers from yesterday's class. You may need to access the documentation to see what the function is actually doing.

```
[6]: # Disjoint sum (disjoint union) of T and R
PdQ = Q.disjoint_union(P)
# Ordinal sum of P and Q
PoQ = P.ordinal_sum(Q)
# Cartesian product of P and Q
PxQ = P.product(Q)
# Ordinal Product of P and Q
PoxQ = P.ordinal_product(Q)
```

```
[7]: g1 = PdQ.plot(
    axes=False,
    element_labels={v: v[1] for v in PdQ}
)

g2 = PoQ.plot(
    axes=False,
```

```

    element_labels={v: v[1] for v in PoQ}
)

g3 = PxQ.plot(
    axes=False,
    element_labels={v: v[1] for v in PxQ}
)

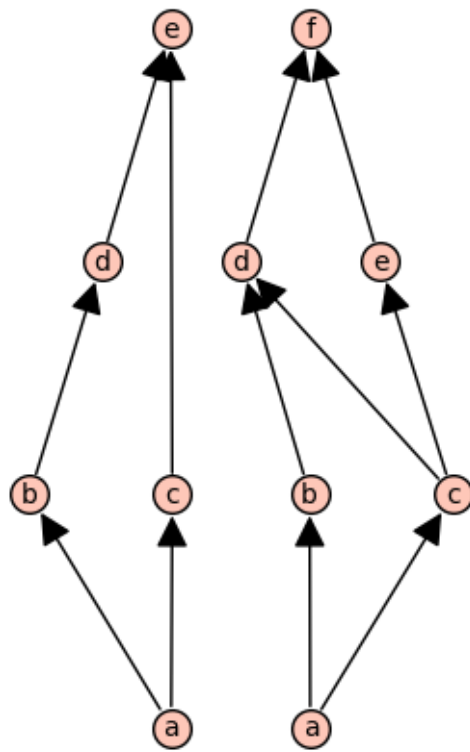
g4 = PoxQ.plot(
    axes=False,
    element_labels={v: v[1] for v in PoxQ}
)

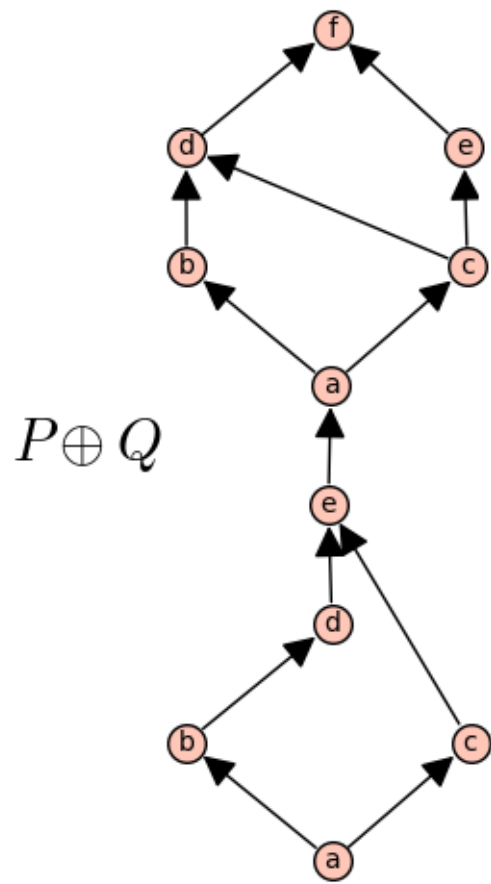
h = PoQ.height() # This is a constant to scale things
g1.set_axes_range(-h, h, -1, h + 0.5)

(g1 + text(r"$P + Q$", (1.5, 3.5),
           fontsize=22, color='black',
           horizontal_alignment='center')).show()
(g2 + text(r"$P \oplus Q$", (1.5, 3.5),
           fontsize=22, color='black',
           horizontal_alignment='center')).show()
(g3 + text(r"$P \times Q$", (1.5, 6),
           fontsize=22, color='black',
           horizontal_alignment='center')).show()
(g4 + text(r"$P \otimes Q$", (1.5, 10),
           fontsize=22, color='black',
           horizontal_alignment='center')).show()

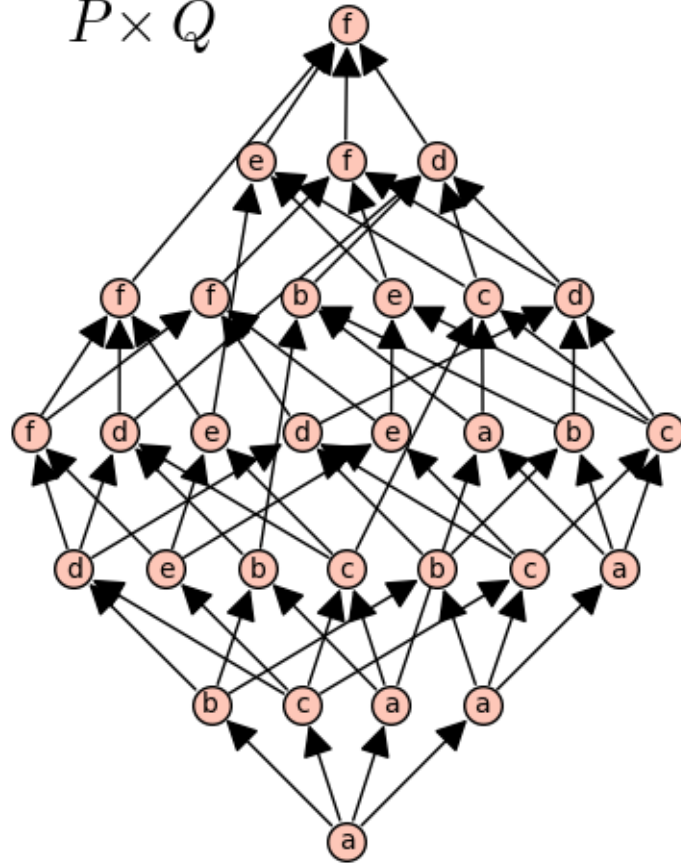
```

$$P + Q$$

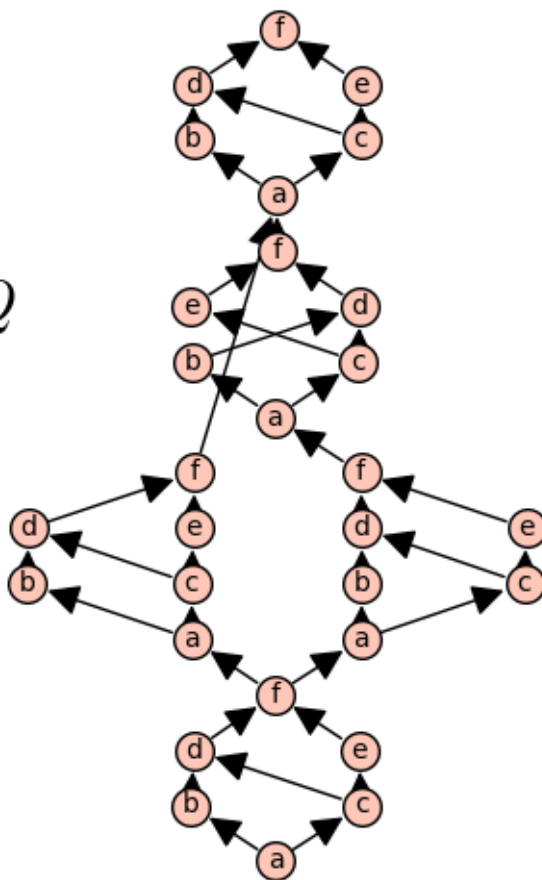




$P \times Q$



$P \otimes Q$



1.3.4 The following function is code for the rank generating function of a poset P . It will not work if your poset does not have a rank function.

Evaluate the cell to define the function and then use it to find the rank generating functions of our example posets.

```
[8]: def rank_generating_function(P):
    if not P.is_ranked():
        return "Does not exist, poset is not ranked"
    f = P.rank_function()
    rk = P.rank()
    list_of_ranks = [f(i) for i in P]
    R.<x> = PolynomialRing(QQ)
    rgf = 0
    for elem in list_of_ranks:
        rgf += x^elem
    return rgf

# Poset dictionary
posets = {
```

```

    "P": P,
    "Q": Q,
    "R": R,
    "S": S,
    "T": T,
    "N": N
}

for key,poset in posets.items():
    print(f"Rank generating function of {key}:␣
↪{rank_generating_function(poset)}")

```

```

Rank generating function of P: Does not exist, poset is not ranked
Rank generating function of Q:  $x^3 + 2x^2 + 2x + 1$ 
Rank generating function of R:  $x + 2$ 
Rank generating function of S:  $2x + 2$ 
Rank generating function of T:  $x^2 + 2x + 1$ 
Rank generating function of N:  $2x + 2$ 

```

1.3.5 An order ideal is a subset of poset elements that is closed downward.

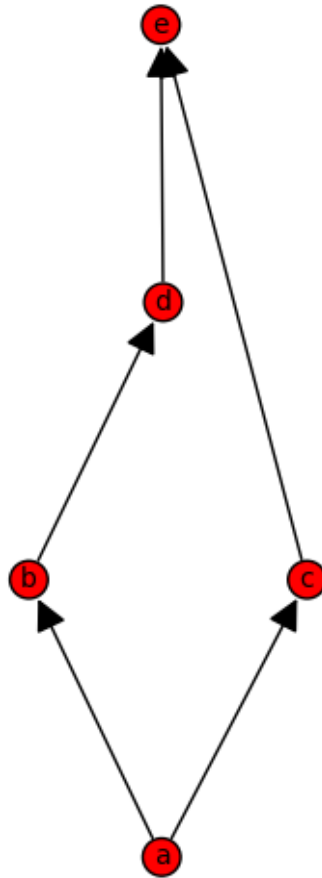
Let's make a random order ideal, using the code below. The green elements are the order ideal. The red elements are an order filter.

```

[9]: oi = P.random_order_ideal()
    list(oi)
    P.order_ideal_plot(oi)

```

[9]:



1.4 Here is one of my favorite posets: the triangle

```

[10]: elemP7 = [11, 12, 13, 14, 15, 16, 17, 21, 22, 23, 24, 25, 26, 31, 32, 33, 34,
↪35, 41, 42, 43, 44, 51, 52, 53, 61, 62, 71]
relP7 = [[11,21],[12, 21],[12, 22],[13, 22],[13, 23],[14, 23],[14,
↪24],[15,24],[15,25],[16,25],[16,26],[17,26],[21,31],[22,31],[22,32],[23,32],[23,33],[24,33]
P7 = Poset((elemP7,relP7))

```

1.5 Evaluate the following cells to see some ways to plot posets

1.5.1 The normal way

```

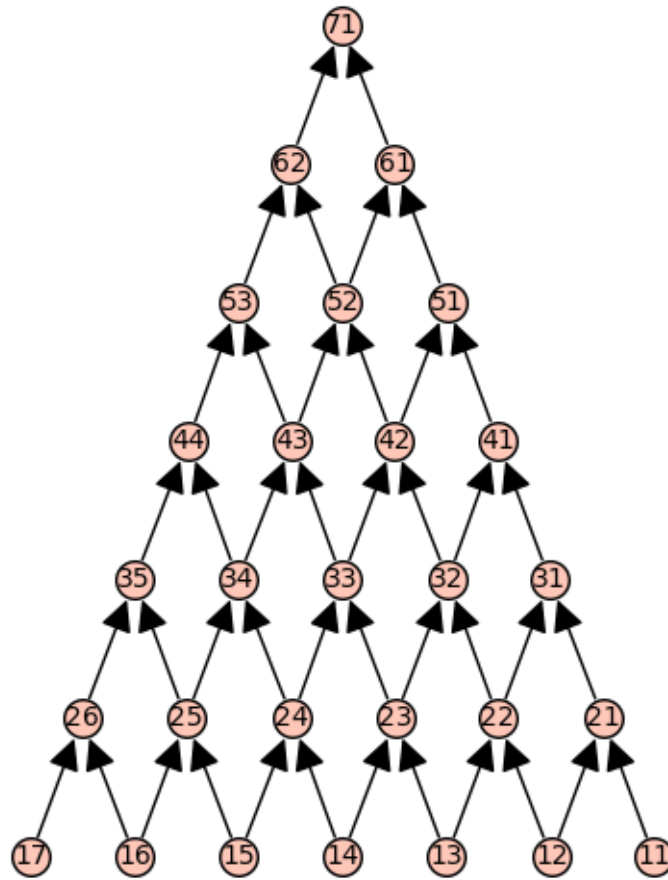
[11]: plot(P7)

```

```

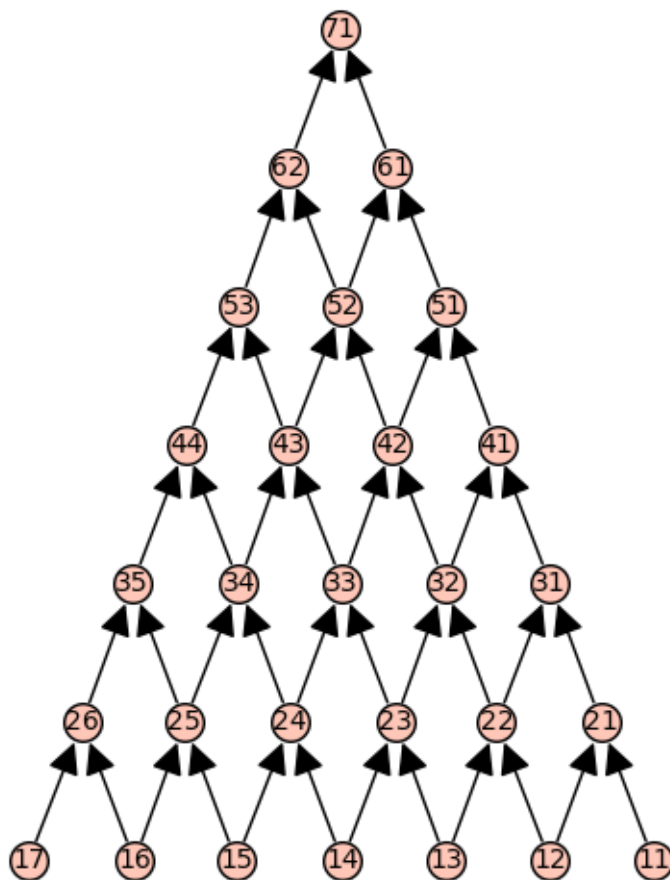
[11]:

```



1.5.2 The pretty way

```
[12]: show(P7.plot())
```



1.5.3 Latex code for putting in your paper

[13]: latex(P7)

Warning: `tikz.sty` is not part of this computer's TeX installation.
This package is required to render graphs in LaTeX.
Visit '...'.

Warning: `tkz-graph.sty` is not part of this computer's TeX installation.
This package is required to render graphs in LaTeX.
Visit '<https://www.ctan.org/pkg/tkz-graph>'.

Warning: `tkz-berge.sty` is not part of this computer's TeX installation.
This package is required to render graphs in LaTeX.
Visit '<https://www.ctan.org/pkg/tkz-berge>'.

```

[13]: \begin{tikzpicture}
\definecolor{cv0}{rgb}{0.0,0.0,0.0}
\definecolor{cfv0}{rgb}{1.0,1.0,1.0}
\definecolor{clv0}{rgb}{0.0,0.0,0.0}
\definecolor{cv1}{rgb}{0.0,0.0,0.0}
\definecolor{cfv1}{rgb}{1.0,1.0,1.0}
\definecolor{clv1}{rgb}{0.0,0.0,0.0}
\definecolor{cv2}{rgb}{0.0,0.0,0.0}
\definecolor{cfv2}{rgb}{1.0,1.0,1.0}
\definecolor{clv2}{rgb}{0.0,0.0,0.0}
\definecolor{cv3}{rgb}{0.0,0.0,0.0}
\definecolor{cfv3}{rgb}{1.0,1.0,1.0}
\definecolor{clv3}{rgb}{0.0,0.0,0.0}
\definecolor{cv4}{rgb}{0.0,0.0,0.0}
\definecolor{cfv4}{rgb}{1.0,1.0,1.0}
\definecolor{clv4}{rgb}{0.0,0.0,0.0}
\definecolor{cv5}{rgb}{0.0,0.0,0.0}
\definecolor{cfv5}{rgb}{1.0,1.0,1.0}
\definecolor{clv5}{rgb}{0.0,0.0,0.0}
\definecolor{cv6}{rgb}{0.0,0.0,0.0}
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\definecolor{clv6}{rgb}{0.0,0.0,0.0}
\definecolor{cv7}{rgb}{0.0,0.0,0.0}
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\definecolor{clv7}{rgb}{0.0,0.0,0.0}
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\definecolor{cv11}{rgb}{0.0,0.0,0.0}
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\definecolor{cv13}{rgb}{0.0,0.0,0.0}
\definecolor{cfv13}{rgb}{1.0,1.0,1.0}
\definecolor{clv13}{rgb}{0.0,0.0,0.0}
\definecolor{cv14}{rgb}{0.0,0.0,0.0}
\definecolor{cfv14}{rgb}{1.0,1.0,1.0}
\definecolor{clv14}{rgb}{0.0,0.0,0.0}
\definecolor{cv15}{rgb}{0.0,0.0,0.0}

```

```

\definecolor{cfv15}{rgb}{1.0,1.0,1.0}
\definecolor{clv15}{rgb}{0.0,0.0,0.0}
\definecolor{cv16}{rgb}{0.0,0.0,0.0}
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\definecolor{cv2v5}{rgb}{0.0,0.0,0.0}
\definecolor{cv3v4}{rgb}{0.0,0.0,0.0}
\definecolor{cv3v7}{rgb}{0.0,0.0,0.0}
\definecolor{cv4v5}{rgb}{0.0,0.0,0.0}
\definecolor{cv4v8}{rgb}{0.0,0.0,0.0}
\definecolor{cv5v9}{rgb}{0.0,0.0,0.0}

```



```

\definecolor{cv6v7}{rgb}{0.0,0.0,0.0}
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\definecolor{cv25v26}{rgb}{0.0,0.0,0.0}
\definecolor{cv26v27}{rgb}{0.0,0.0,0.0}
%
\Vertex[style={minimum size=1.0cm,draw=cv0,fill=cfv0,text=clv0,shape=circle},LabelOut=false,L=\hbox{$17$},x=0.0878cm,y=5.0cm]{v0}
\Vertex[style={minimum size=1.0cm,draw=cv1,fill=cfv1,text=clv1,shape=circle},LabelOut=false,L=\hbox{$16$},x=0.3667cm,y=3.919cm]{v1}
\Vertex[style={minimum size=1.0cm,draw=cv2,fill=cfv2,text=clv2,shape=circle},LabelOut=false,L=\hbox{$26$},x=0.1712cm,y=4.2869cm]{v2}
\Vertex[style={minimum size=1.0cm,draw=cv3,fill=cfv3,text=clv3,shape=circle},LabelOut=false,L=\hbox{$15$},x=1.0844cm,y=2.9208cm]{v3}
\Vertex[style={minimum size=1.0cm,draw=cv4,fill=cfv4,text=clv4,shape=circle},LabelOut=false,L=\hbox{$25$},x=0.5199cm,y=3.1963cm]{v4}
\Vertex[style={minimum size=1.0cm,draw=cv5,fill=cfv5,text=clv5,shape=circle},LabelOut=false,L=\hbox{$35$},x=0.0089cm,y=3.4505cm]{v5}
\Vertex[style={minimum size=1.0cm,draw=cv6,fill=cfv6,text=clv6,shape=circle},LabelOut=false,L=\hbox{$36$},x=0.0089cm,y=3.4505cm]{v6}

```

```

elOut=false,L=\hbox{$14$},x=2.0706cm,y=2.3436cm]{v6}
\Vertex[style={minimum size=1.0cm,draw=cv7,fill=cfv7,text=clv7,shape=circle},LabelOut=false,L=\hbox{$24$},x=1.3594cm,y=2.3354cm]{v7}
\Vertex[style={minimum size=1.0cm,draw=cv8,fill=cfv8,text=clv8,shape=circle},LabelOut=false,L=\hbox{$34$},x=0.6236cm,y=2.3394cm]{v8}
\Vertex[style={minimum size=1.0cm,draw=cv9,fill=cfv9,text=clv9,shape=circle},LabelOut=false,L=\hbox{$44$},x=0.0cm,y=2.5066cm]{v9}
\Vertex[style={minimum size=1.0cm,draw=cv10,fill=cfv10,text=clv10,shape=circle},LabelOut=false,L=\hbox{$13$},x=3.1285cm,y=1.8055cm]{v10}
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\Vertex[style={minimum size=1.0cm,draw=cv12,fill=cfv12,text=clv12,shape=circle},LabelOut=false,L=\hbox{$33$},x=1.6164cm,y=1.6148cm]{v12}
\Vertex[style={minimum size=1.0cm,draw=cv13,fill=cfv13,text=clv13,shape=circle},LabelOut=false,L=\hbox{$43$},x=0.7753cm,y=1.5234cm]{v13}
\Vertex[style={minimum size=1.0cm,draw=cv14,fill=cfv14,text=clv14,shape=circle},LabelOut=false,L=\hbox{$53$},x=0.105cm,y=1.611cm]{v14}
\Vertex[style={minimum size=1.0cm,draw=cv15,fill=cfv15,text=clv15,shape=circle},LabelOut=false,L=\hbox{$12$},x=4.211cm,y=1.1787cm]{v15}
\Vertex[style={minimum size=1.0cm,draw=cv16,fill=cfv16,text=clv16,shape=circle},LabelOut=false,L=\hbox{$22$},x=3.4682cm,y=1.2236cm]{v16}
\Vertex[style={minimum size=1.0cm,draw=cv17,fill=cfv17,text=clv17,shape=circle},LabelOut=false,L=\hbox{$32$},x=2.6833cm,y=1.063cm]{v17}
\Vertex[style={minimum size=1.0cm,draw=cv18,fill=cfv18,text=clv18,shape=circle},LabelOut=false,L=\hbox{$42$},x=1.8083cm,y=0.8651cm]{v18}
\Vertex[style={minimum size=1.0cm,draw=cv19,fill=cfv19,text=clv19,shape=circle},LabelOut=false,L=\hbox{$52$},x=0.9971cm,y=0.7162cm]{v19}
\Vertex[style={minimum size=1.0cm,draw=cv20,fill=cfv20,text=clv20,shape=circle},LabelOut=false,L=\hbox{$62$},x=0.2664cm,y=0.7578cm]{v20}
\Vertex[style={minimum size=1.0cm,draw=cv21,fill=cfv21,text=clv21,shape=circle},LabelOut=false,L=\hbox{$11$},x=5.0cm,y=0.537cm]{v21}
\Vertex[style={minimum size=1.0cm,draw=cv22,fill=cfv22,text=clv22,shape=circle},LabelOut=false,L=\hbox{$21$},x=4.3803cm,y=0.6666cm]{v22}
\Vertex[style={minimum size=1.0cm,draw=cv23,fill=cfv23,text=clv23,shape=circle},LabelOut=false,L=\hbox{$31$},x=3.6041cm,y=0.5943cm]{v23}
\Vertex[style={minimum size=1.0cm,draw=cv24,fill=cfv24,text=clv24,shape=circle},LabelOut=false,L=\hbox{$41$},x=2.7955cm,y=0.3894cm]{v24}
\Vertex[style={minimum size=1.0cm,draw=cv25,fill=cfv25,text=clv25,shape=circle},LabelOut=false,L=\hbox{$51$},x=1.9527cm,y=0.1224cm]{v25}
\Vertex[style={minimum size=1.0cm,draw=cv26,fill=cfv26,text=clv26,shape=circle},LabelOut=false,L=\hbox{$61$},x=1.1646cm,y=0.0cm]{v26}
\Vertex[style={minimum size=1.0cm,draw=cv27,fill=cfv27,text=clv27,shape=circle},LabelOut=false,L=\hbox{$71$},x=0.4806cm,y=0.0012cm]{v27}
%
\Edge[lw=0.1cm,style={post, bend right,color=cv0v2,},](v0)(v2)
\Edge[lw=0.1cm,style={post, bend right,color=cv1v2,},](v1)(v2)
\Edge[lw=0.1cm,style={post, bend right,color=cv1v4,},](v1)(v4)

```

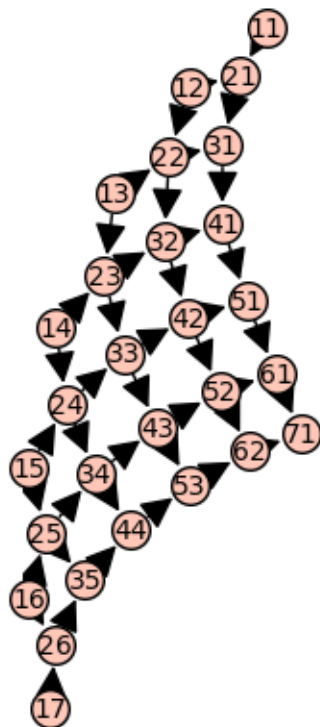
```

\Edge[lw=0.1cm,style={post, bend right,color=cv2v5,},](v2)(v5)
\Edge[lw=0.1cm,style={post, bend right,color=cv3v4,},](v3)(v4)
\Edge[lw=0.1cm,style={post, bend right,color=cv3v7,},](v3)(v7)
\Edge[lw=0.1cm,style={post, bend right,color=cv4v5,},](v4)(v5)
\Edge[lw=0.1cm,style={post, bend right,color=cv4v8,},](v4)(v8)
\Edge[lw=0.1cm,style={post, bend right,color=cv5v9,},](v5)(v9)
\Edge[lw=0.1cm,style={post, bend right,color=cv6v7,},](v6)(v7)
\Edge[lw=0.1cm,style={post, bend right,color=cv6v11,},](v6)(v11)
\Edge[lw=0.1cm,style={post, bend right,color=cv7v8,},](v7)(v8)
\Edge[lw=0.1cm,style={post, bend right,color=cv7v12,},](v7)(v12)
\Edge[lw=0.1cm,style={post, bend right,color=cv8v9,},](v8)(v9)
\Edge[lw=0.1cm,style={post, bend right,color=cv8v13,},](v8)(v13)
\Edge[lw=0.1cm,style={post, bend right,color=cv9v14,},](v9)(v14)
\Edge[lw=0.1cm,style={post, bend right,color=cv10v16,},](v10)(v16)
\Edge[lw=0.1cm,style={post, bend right,color=cv10v11,},](v10)(v11)
\Edge[lw=0.1cm,style={post, bend right,color=cv11v17,},](v11)(v17)
\Edge[lw=0.1cm,style={post, bend right,color=cv11v12,},](v11)(v12)
\Edge[lw=0.1cm,style={post, bend right,color=cv12v18,},](v12)(v18)
\Edge[lw=0.1cm,style={post, bend right,color=cv12v13,},](v12)(v13)
\Edge[lw=0.1cm,style={post, bend right,color=cv13v19,},](v13)(v19)
\Edge[lw=0.1cm,style={post, bend right,color=cv13v14,},](v13)(v14)
\Edge[lw=0.1cm,style={post, bend right,color=cv14v20,},](v14)(v20)
\Edge[lw=0.1cm,style={post, bend right,color=cv15v16,},](v15)(v16)
\Edge[lw=0.1cm,style={post, bend right,color=cv15v22,},](v15)(v22)
\Edge[lw=0.1cm,style={post, bend right,color=cv16v17,},](v16)(v17)
\Edge[lw=0.1cm,style={post, bend right,color=cv16v23,},](v16)(v23)
\Edge[lw=0.1cm,style={post, bend right,color=cv17v18,},](v17)(v18)
\Edge[lw=0.1cm,style={post, bend right,color=cv17v24,},](v17)(v24)
\Edge[lw=0.1cm,style={post, bend right,color=cv18v19,},](v18)(v19)
\Edge[lw=0.1cm,style={post, bend right,color=cv18v25,},](v18)(v25)
\Edge[lw=0.1cm,style={post, bend right,color=cv19v20,},](v19)(v20)
\Edge[lw=0.1cm,style={post, bend right,color=cv19v26,},](v19)(v26)
\Edge[lw=0.1cm,style={post, bend right,color=cv20v27,},](v20)(v27)
\Edge[lw=0.1cm,style={post, bend right,color=cv21v22,},](v21)(v22)
\Edge[lw=0.1cm,style={post, bend right,color=cv22v23,},](v22)(v23)
\Edge[lw=0.1cm,style={post, bend right,color=cv23v24,},](v23)(v24)
\Edge[lw=0.1cm,style={post, bend right,color=cv24v25,},](v24)(v25)
\Edge[lw=0.1cm,style={post, bend right,color=cv25v26,},](v25)(v26)
\Edge[lw=0.1cm,style={post, bend right,color=cv26v27,},](v26)(v27)
%
\end{tikzpicture}

```

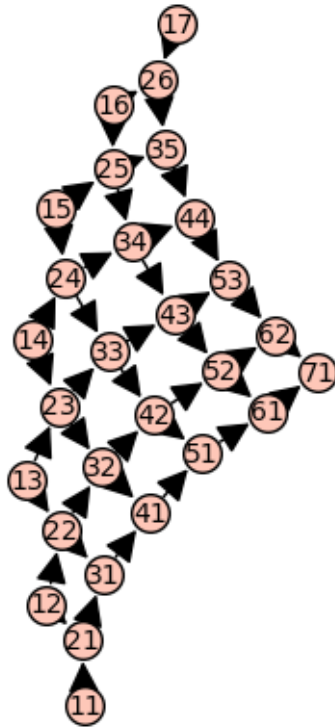
1.5.4 The boingy way

```
[14]: H7 = P7.hasse_diagram()  
      show(H7)
```



1.5.5 The sticky way

```
[15]: H7.plot(save_pos=True)  
      show(H7)
```



1.5.6 Exercise 3: Use these various plot methods on our example posets. Which plot method is your favorite?

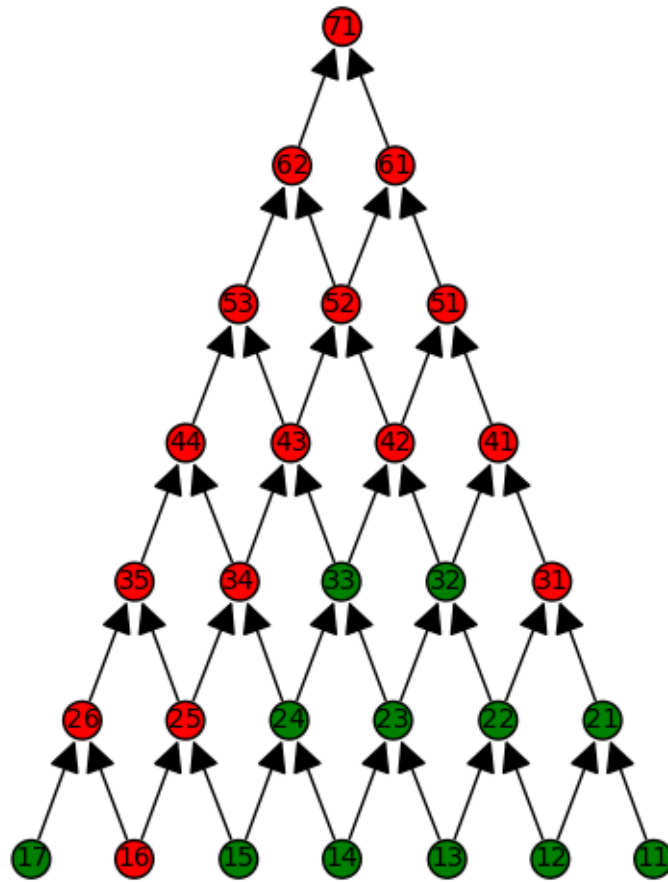
[16]: I like boingy

```
Cell In[16], line 1
  I like boingy
  ^
SyntaxError: invalid syntax
```

1.5.7 The reason the triangle poset is one of my favorite posets is that order ideals of the triangle poset are Dyck paths!

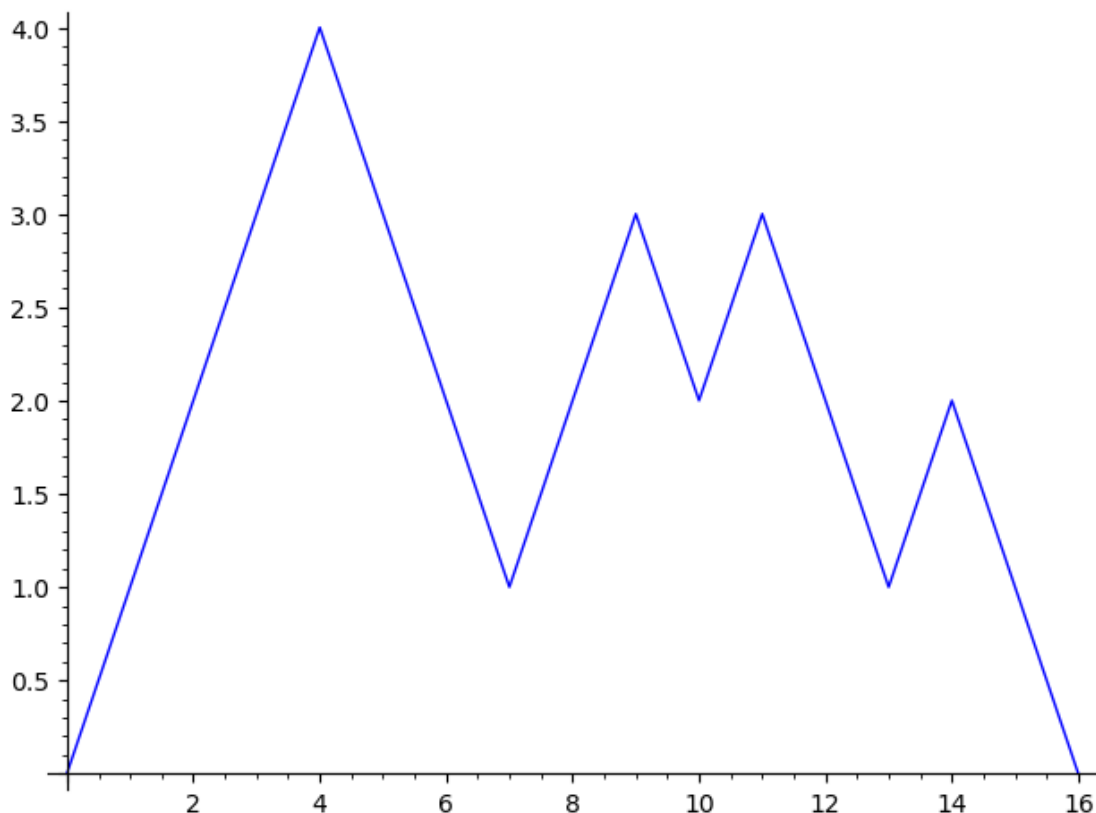
```
[ ]: P7oi = P7.random_order_ideal()
      P7.order_ideal_plot(P7oi)
```

[]:



```
[ ]: d = DyckWords(8).random_element()
plot(d)
```

```
[ ]:
```



1.6 We can use Sage to construct new posets from old

1.6.1 Exercise 4: Using our example posets P,Q,R,S,T,N, construct new posets as examples of the disjoint sum, ordinal sum, and direct (cartesian) product. Plot the new posets you create.

```
[ ]: # Poset dictionary
PO = {
    "P": P,
    "Q": Q,
    "R": R,
    "S": S,
    "T": T,
    "N": N
}

"""
examples of poset operations:

# Disjoint sum (disjoint union) of T and R
PdQ = Q.disjoint_union(P)
```

```

# Ordinal sum of P and Q
PoQ = P.ordinal_sum(Q)
# Cartesian product of P and Q
PxQ = P.product(Q)
# Ordinal Product of P and Q
PoXQ = P.ordinal_product(Q)
"""

RdT = R.disjoint_union(T)
RoN = R.ordinal_sum(N)
NxR = N.product(R)
NoxR = N.ordinal_product(R)

```

```

[18]: g1 = RdT.plot(
        axes=False,
        element_labels={v: v[1] for v in RdT}
    )

g2 = RoN.plot(
        axes=False,
        element_labels={v: v[1] for v in RoN}
    )

g3 = NxR.plot(
        axes=False,
        element_labels={v: v[1] for v in NxR}
    )

g4 = NoxR.plot(
        axes=False,
        element_labels={v: v[1] for v in NoxR}
    )

h = PoQ.height() # This is a constant to scale things
g1.set_axes_range(-h, h, -1, h + 0.5)

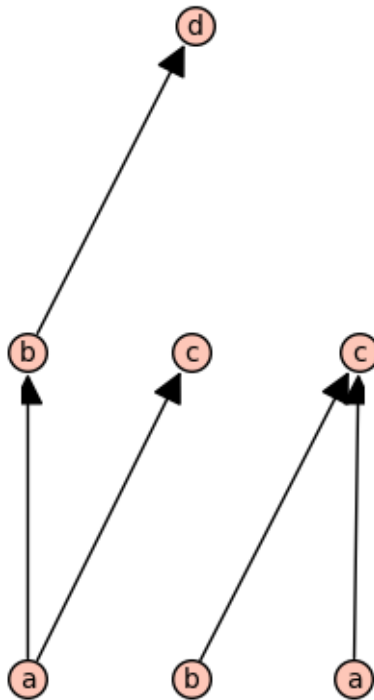
(g1 + text(r"$R + T$", (1, 2.5),
           fontsize=22, color='black',
           horizontal_alignment='center')).show()
(g2 + text(r"$R \oplus N$", (1.5, 3.5),
           fontsize=22, color='black',
           horizontal_alignment='center')).show()
(g3 + text(r"$N \times R$", (1, 2.5),
           fontsize=22, color='black',
           horizontal_alignment='center')).show()
(g4 + text(r"$N \otimes R$", (1.5, 3.5),

```

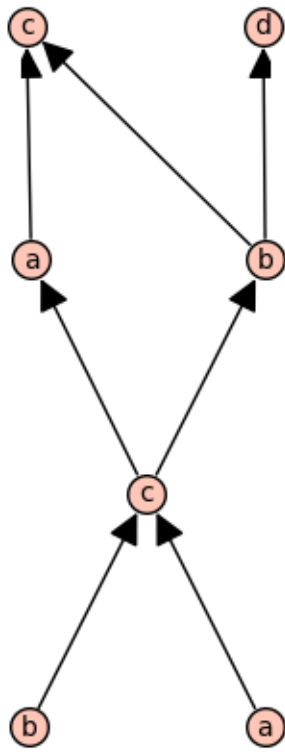


```
fontsize=22, color='black',  
horizontal_alignment='center')).show()
```

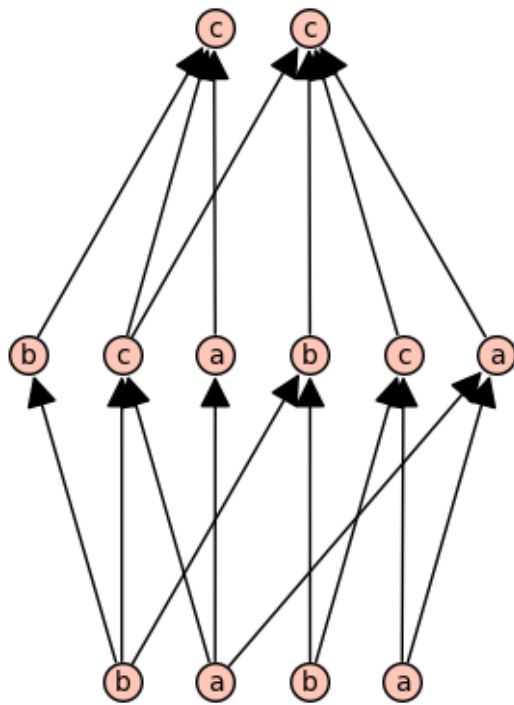
$$R + T$$



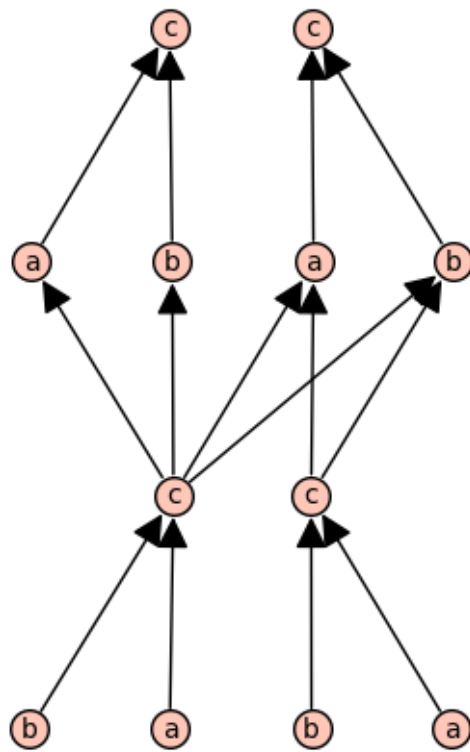
$$R \oplus N$$



$$N \times R$$



$$N \otimes R$$



[0] :